

SAT UZBEKISTAN

Equivalent questions ask us to identify which of the answer choices is equivalent to a given expression. These questions often look difficult, but they are easy as long as we use our Equivalent Trick or Desmos Equivalent Hack!

Equivalent Trick

1. Plug in $x = 1$ to the original expression and solve.
2. Plug in $x = 1$ to the answer choices and solve.
3. Find which answer choice gives you the same value as the original expression. This is the correct answer. Incorrect answer choices give you a value that is not the same as to the original expression.
4. If you cannot tell which answer choice is correct after plugging in $x = 1$ because multiple answer choices equal the same value, plug in $x = 2$ or another simple value for x .

If you do the math correctly, you can identify the correct answer for any equivalent questions without doing any fancy algebra!

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Let's see how this works with some examples. To begin, we will start with an easier example.

Example 1: Which of the following expressions is equivalent to $8x^3 - 4x$?

- A) $2x(4x^2 - 2)$ B) $-2x^2(2x - 1)$ C) $8x(x^2 - 1)$ D) $2x(2x - 2)^2$

Solution: Method #1 – Equivalent Trick: First, we plug in $x = 1$ to the expression $8x^3 - 4x$ and solve.

$$8(1)^3 - 4(1) = 8 - 4 = 4$$

We know that if $x = 1$, the expression equals 4. The equivalent answer choice must also equal 4. To find which answer choice is correct, plug in $x = 1$ to the answer choices and find which one equals 4.

For answer choice A: $2(1)(4(1)^2 - 2) \rightarrow 2(4 - 2) = 4$ (A is correct)

For answer choice B: $-2(1)^2(2(1) - 1) \rightarrow -2(2 - 1) = -2$ (B is incorrect)

For answer choice C: $8(1)((1)^2 - 1) \rightarrow 8(1 - 1) = 0$ (C is incorrect)

For answer choice D: $2(1)(2(1) - 2)^2 \rightarrow 2(2 - 2)^2 = 0$ (D is incorrect)

When we plug in $x = 1$ to the answer choices and solve, only answer choice A gives us 4. Since both the original expression and the expression in answer choice A give us the same value, they must be equivalent! The answer is A.

1. Which of the following is equivalent to $x^2 - 5x - 18$?

- A) $(x + 3)(x - 6) - 2x$
B) $(x + 3)(x - 6) + 5x$
C) $(x + 2)(x - 9) - 2x$
D) $(x + 2)(x - 9) + 5x$

2. Which of the following expressions is equivalent to $18x^2 + 6x$?

- A) $2x(18x + 3)$
B) $6x(3x + 6)$
C) $18x(x + 1)$
D) $3x(6x + 2)$

3. Which of the following is equivalent to the expression below?

$$2(3x + 1)(2x + 2)$$

- A) $8x + 3$
B) $12x^2 + 4$
C) $6x^2 + 8x + 2$
D) $12x^2 + 16x + 4$

4. Which of the following expressions is an equivalent form of the expression $(2x + 2)^2 + (2x + 2)$?

- A) $2x^2 + 10x + 6$
B) $4x^2 + 16x + 10$
C) $(2x + 2)(2x + 3)$
D) $(4x + 8)(x + 1)$

5. Which of the following expressions is an equivalent form of $x^2 + 10x + 20$?

- A) $(x + 4)(x + 6) - 4$
B) $(x + 4)(x + 6) + 4x$
C) $(x + 12)(x - 2) + 4$
D) $(x + 12)(x - 2) + 24$

6. $\frac{2}{x+7} + \frac{1}{2}$

Which of the following expressions is equivalent to the one above where $x \neq -7$?

- A) $\frac{x+11}{2x+14}$
B) $\frac{x+7}{2x+14}$
C) $\frac{5}{x+7}$
D) $\frac{3}{x+9}$

7. Which of the following is equivalent to the expression below?

$$x^2 + 8x + 11$$

- A) $(x + 4)^2 - 5$
B) $(x + 4)^2 + 3$
C) $(x + 4)^2 + 11$
D) $(x + 4)^2$

8. Which of the following is equivalent to $(-3x + 4)^2 - 9x^2$?

- A) $-16(x + 1)$
B) $4(-2x + 4)$
C) $-8(3x - 2)$
D) $8(x + 2)$

9. Which of the following expressions is equivalent to $x + 2 + \frac{3}{x+1}$ where $x \neq -1$? (2)

A) $\frac{x+5}{x+1}$
 B) $\frac{x^2+3x+5}{x+1}$
 C) $\frac{x^2+5x}{x+1}$
 D) $x + 3$

10. For $a > 0$, which of the following is equivalent to $\frac{6a^2-3a}{2a+1}$? (2)

A) $3a$
 B) $3a - 3$
 C) $3a - 3 + \frac{3}{2a+1}$
 D) $3a - \frac{a}{2a+1}$

11. Which of the following is equivalent to the expression below? (2)

$$16x^4 + 8x^3 - 24x^2 - 12x$$

A) $(2x + 1)(2x^2 - 3)$
 B) $x(4x + 2)(4x^2 - 6)$
 C) $4x(2x - 1)(2x^2 - 3)$
 D) $4x(2x + 3)(2x^2 + 1)$

12. Which expression is equivalent to $5x^6y^3 + 20x^4y^3$? (3)

A) $5x^4y^3(x^2 + 4)$
 B) $5x^4y^3(x^{\frac{3}{2}} + 4)$
 C) $5x^4y^3(4x^2)$
 D) $5x^4y^3(4x^4)$

13. Which expression is equivalent to $\frac{7x(x-9)-6(x-9)}{3x-27}$, where $x > 9$? (3)

A) $\frac{x-9}{3}$
 B) $\frac{7x^2-69x-54}{3x-27}$
 C) $\frac{7x^2-69x+54}{x-9}$
 D) $\frac{7x-6}{3}$

14. Which expression is equivalent to $\frac{3+4x}{81-256x^4}$, where $x > 1$? (3)

A) $\frac{1}{27-64x^3}$
 B) $27 - 64x^3$
 C) $\frac{1}{(9+16x^2)(3-4x)}$
 D) $(9 + 16x^2)(3 - 4x)$

15. For $x \neq 4$, which of the following expressions is equivalent to $\frac{y+7}{x-4} + \frac{y(x-4)}{x^2y-4xy}$? (4)

A) $\frac{xy+y+3}{x^3-4xy^2+16xy}$
 B) $\frac{xy^2+6xy-4y}{x^2y-4xy}$
 C) $\frac{xy^2+8xy-4y}{x^2y-4xy}$
 D) $\frac{xy^2+8xy-4y}{x^3y-8x^2y+16xy}$